

MOHAN SILAMBARASU ELANGKUMARAN

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EDUCATION

George Mason University

Fairfax, VA

Master of Science in Information Technology,

Aug 2024 – May 2026

Specialization: Data Analytics and Intelligence Methods | **GPA: 4.0** | **Academic Excellence Award Recipient**

Coursework: Natural Language Processing, Interpretable Machine Learning, Big Data Analytics, Cloud Computing, Distributed Systems, Database Management Systems, Data Structures and Algorithms, Object-Oriented Programming, Computing Platforms, Operating Systems, Networking & Architecture.

EXPERIENCE

Solvative

Ahmedabad, India

Frontend Developer

Apr 2022 – Jul 2024

- Architected real-time, data-intensive React applications for IIoT platforms processing live sensor data streams for U.S. manufacturing clients, reducing alert response time by 58% via optimized virtual DOM rendering, efficient component-level state management, targeted performance profiling.
- Engineered scalable, reusable React component libraries with TypeScript strict mode, enforcing type safety, separation of concerns, and modular architecture patterns that reduced cross-team onboarding time and improved long-term codebase maintainability across multiple product lines.
- Integrated React frontends with microservices and backend systems via RESTful APIs and JSON-based data contracts, owning the full API consumption layer from asynchronous request handling and error boundaries to UI state synchronization, reducing frontend-backend integration bugs reported in QA.
- Accelerated feature delivery and code quality by adopting AI-assisted development workflows with GitHub Copilot and Cursor, reducing boilerplate generation time and maintaining consistent, reviewable output across Agile sprint cycles in a fast-paced production environment.
- Streamlined production releases and deployment reliability by managing GitLab CI/CD pipelines, conducting structured code reviews, collaborating with backend engineers, QA teams, and product managers across Agile/Scrum ceremonies in a cross-functional delivery environment.

Kantar Analytics Practice

Bengaluru, India

Senior Analyst – Data Strategy and Engineering

May 2021 – Feb 2022

- Automated end-to-end enterprise data workflows using Python, pandas, and SQL Server, improving data integrity to 99.9% across reporting pipelines and eliminating 70% of manual processing effort, saving 2 to 3 weeks per delivery cycle.
- Designed and delivered API-ready data transformation, validation, and preprocessing pipelines in Python, enabling reliable structured data feeds consumed by downstream frontend analytics dashboards and business intelligence reporting layers.
- Optimized complex multi-join SQL queries against large-scale enterprise datasets, improving backend data preparation performance and ensuring consistent, accurate data flows available to frontend and reporting consumers.
- Delivered production-ready data pipeline solutions in Agile delivery cycles using Azure DevOps, translating ambiguous business requirements into scalable technical implementations and maintaining predictable sprint velocity across a cross-functional analyst and engineering team.

PROJECTS

SupportAI | Multi-Agent AI Customer Support System | React, TypeScript, Node.js, Express.js, Groq LLM, REST APIs, LLM Integration

github.com/mohansilambarasu/ai-support-agent

- Architected a production-grade React/TypeScript frontend for a three-agent sequential LLM pipeline with real-time agent reasoning trace visualization, dynamically rendering classification decisions, knowledge base matches, and confidence scores as each agent executes.
- Engineered a human-in-the-loop review dashboard with editable AI-generated reply workflows, approve/reject controls, and live agent status indicators, integrating LLM backend APIs via REST to deliver a complete AI-powered interface with full human oversight of outbound communications.

StudyAI | RAG-Powered Document Intelligence System | React, TypeScript, FastAPI, LangChain LCEL, Pinecone, HuggingFace, Groq

LLM, REST APIs - github.com/mohansilambarasu/ai-study-assistant

- Built a production-ready React/TypeScript conversational UI with drag-and-drop PDF upload, per-message source citation rendering, and persistent multi-turn memory, consuming a FastAPI RAG backend via REST APIs to deliver a seamless AI-native user experience.
- Integrated LangChain LCEL, Pinecone vector database semantic search, HuggingFace embeddings, and Groq LLM inference through a clean API abstraction layer, demonstrating full-stack ownership from React UI to AI inference with per-session document isolation and prompt-engineered hallucination prevention.

MERN Stack Task and Content Management Application | MongoDB, Express.js, React.js, Node.js, REST APIs, JavaScript, Component-Based Architecture

- Built a full-stack MERN application with dynamic React views, reusable component architecture, RESTful API communication, middleware handling, and MongoDB data modeling, demonstrating complete full-stack ownership across the entire frontend to backend request lifecycle.

Java Shortest Path Finder and Graph Algorithm System Java, Object-Oriented Programming, Data Structures, Algorithms, Graph Theory

- Implemented Dijkstra's shortest path and BFS/DFS graph traversal algorithms in Java using OOP design principles and core data structures including adjacency lists, priority queues, and hash maps, demonstrating strong CS fundamentals in algorithmic problem solving, time and space complexity analysis, and maintainable system design.

SKILLS

Frontend: React.js, TypeScript, JavaScript ES6+, HTML5, CSS3, Tailwind CSS, React Router, Context API, Redux, Responsive Design, Component-Based Architecture, Micro-Frontend Patterns, Web Performance Optimization, Accessibility (WCAG), Cross-Browser Compatibility

AI / LLM: LangChain, LangChain LCEL, LangGraph, OpenAI API, Groq, HuggingFace Transformers, Pinecone, Retrieval-Augmented Generation, Multi-Agent Systems, Vector Embeddings, Prompt Engineering, AI-Powered UI Development, Conversational UI, Streaming API Integration

AI-Augmented Development Tools: GitHub Copilot, Cursor, AI-assisted code generation, Prompt Engineering for development workflows

Backend: Node.js, Express.js, FastAPI, Python, REST APIs, JSON, API Design, Microservices Integration, SQL, pandas

Cloud and Databases: AWS EC2, AWS S3, AWS RDS, MongoDB, PostgreSQL, MySQL, SQLite, Pinecone Serverless

Software Engineering Practices: Data Structures and Algorithms, Object-Oriented Programming, System Design, Code Review, Debugging, Performance Profiling, Git, GitLab CI/CD, Azure DevOps, Agile/Scrum, Postman, Vercel, Jupyter Notebook